

A rectangular array of numbers, symbols or characters is called Matrix. These are identified by their order.

The order of the matrices is given in the form of rows $\times$ number of columns.

A matrix is represented as $[P]_{m \times n}$.

example

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2} \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3 \times 3}$$

Operations

① Addition

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$$

$$\begin{aligned} A+B &= \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix} = \begin{bmatrix} 1+2 & 2+3 \\ 4+6 & 5+7 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 5 \\ 10 & 12 \end{bmatrix} \end{aligned}$$

Subtraction

$$A = \begin{bmatrix} 7 & 2 \\ 4 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 1 \\ 2 & 3 \end{bmatrix}$$

$$A - B = \begin{bmatrix} 7 & 2 \\ 4 & 5 \end{bmatrix} - \begin{bmatrix} 5 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 7-5 & 2-1 \\ 4-2 & 5-3 \end{bmatrix} \\ = \begin{bmatrix} 2 & 1 \\ 2 & 2 \end{bmatrix}$$

Scalar Multiply

refers to the multiplication of each term with the scalar term.

Example Multiply 3 with $\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$

$$3[A] = \begin{bmatrix} 3 \times 1 & 3 \times 2 \\ 3 \times 4 & 3 \times 5 \end{bmatrix}$$

$$\Rightarrow 3[A] = \begin{bmatrix} 3 & 6 \\ 12 & 15 \end{bmatrix}$$

Multiplication of Matrices

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} \times \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 2 + 2 \times 6 & 1 \times 3 + 2 \times 7 \\ 4 \times 2 + 5 \times 6 & 4 \times 3 + 5 \times 7 \end{bmatrix}$$

$$AB = \begin{bmatrix} 14 & 17 \\ 38 & 47 \end{bmatrix}$$

Transpose of Matrix

$$A = \begin{bmatrix} 18 & 17 \\ 38 & 47 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 18 & 38 \\ 17 & 47 \end{bmatrix}$$

Properties of Matrix

- $A + B = B + A$ (Commutative)
- $(A + B) + C = A + (B + C)$ (Associative)
- $AB \neq BA$ (Not commutative)
- $(AB)C = A(BC)$ (Associative)
- $A(B + C) = AB + AC$ (Distributive)

Trace of Matrix

It is the sum of the diagonal elements of a square matrix.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\text{Trace } A = 1 + 5 + 9 = 15$$

Types of Matrices

- Row Matrix: A matrix in which there is only one row.

ex $A = [p \ q \ r]_{1 \times 3}$

- Column Matrix: A matrix in which there is only one column.

ex $A = \begin{bmatrix} p \\ q \\ r \end{bmatrix}_{3 \times 1}$

- Rectangular Matrix: A matrix that doesn't have equal number of rows and columns.

ex $\begin{bmatrix} 1 & 2 & 4 \\ 3 & 6 & 9 \end{bmatrix}_{2 \times 3}$

- Square Matrix: A matrix in which number of rows is equal to number of columns.

ex $= \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 4 & 9 \end{bmatrix}_{3 \times 3}$

- Diagonal Matrix :- A square matrix in which the non-diagonal elements are zero.

ex
$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}_{3 \times 3}$$

- Zero / Null Matrix :- A matrix whose all elements are zero.

ex
$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$$

- Unit / Identity matrix :- A diagonal matrix whose all diagonal elements are 1. It is represented by I .

ex
$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

- Symmetric Matrix :- A square matrix whose transpose is equal to the original matrix. i.e., $A^T = A$

ex

$$A = \begin{bmatrix} 7 & 38 \\ 38 & 7 \end{bmatrix}$$

Here $A = A^T$

$$A^T = \begin{bmatrix} 7 & 38 \\ 38 & 7 \end{bmatrix}$$

- Skew-Symmetric Matrix: A square matrix whose transpose equals to its negative i.e., $A^T = -A$. It is also called antisymmetric or antimetric [1].

ex

$$A = \begin{bmatrix} 0 & 3 & -8 \\ -3 & 0 & 6 \\ 8 & -6 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 0 & -3 & 8 \\ 3 & 0 & -6 \\ -8 & 6 & 0 \end{bmatrix}$$

NOTE If matrix A is a square matrix then $(A - A^t)$ is always skew-symmetric.

- Idempotent Matrix: A matrix is said to be idempotent if $A^2 = A$.

$$\text{ex} \\ \underline{\underline{2}} \quad A = \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix}_{2 \times 2}$$

- Involutory Matrix: A matrix is said to be involutory if $A^2 = I$

$$\text{ex} \\ \underline{\underline{2}} \quad A = \begin{bmatrix} 2 & 1 \\ -3 & -2 \end{bmatrix}$$

- Upper Triangular Matrix: A square matrix in which all the elements below the principal diagonal are 0.

$$\text{ex} \\ \underline{\underline{3}} \quad A = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

- Lower Triangular Matrix: A square matrix in which all the elements above the principal diagonal are 0.

$$\text{ex} \\ \underline{\underline{3}} \quad A = \begin{bmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{bmatrix}$$

• Singular Matrix: A square

matrix whose determinant is 0.

i.e., $|A| = 0$ and is not invertible.

ex $A = \begin{bmatrix} 2 & 2 & 4 \\ 1 & 1 & 2 \\ 3 & 7 & 9 \end{bmatrix}$

Non-Singular Matrix: A square

matrix whose determinant is not zero. i.e., $|A| \neq 0$. Its inverse is defined.

ex $A = \begin{bmatrix} 1 & 2 & -3 \\ 6 & 0 & 8 \\ -1 & 4 & 0 \end{bmatrix}$